COURSE 2 - (AI Agents and Multi-Agent Systems)

1. Introduction to AI Agents

Definition: An AI agent is a system that uses an AI model, instructions, and possibly tools to perform tasks and work toward a goal.
Unlike a simple question-answering system, an AI agent can take actions, use external tools, and coordinate multiple steps to accomplish a task.
Example from the lecture: A purchasing concierge system that helps users order burgers or pizzas by coordinating specialized AI agents.
Key Point: AI agents can perform tasks using reasoning, tools, and interactions with other systems.

2. Evolution of Software Interaction

The lecture discusses a shift from relying on a single, large AI system toward using multiple specialized agents that cooperate.
Monolithic AI System: A single system handles many responsibilities.
Multi-Agent System: Multiple specialized agents handle different responsibilities and communicate to complete a larger task.
Lecture Example:

- A Burger Agent handles burger-related orders.
- A Pizza Agent handles pizza-related orders.
- A Route Agent directs requests to the appropriate agent.

Key Point: Specialized agents divide responsibilities instead of requiring one agent to handle everything.

3. Modularity in AI Systems

Definition: Modularity means dividing a large system into smaller components with distinct responsibilities.
Each component can be developed, maintained, or improved separately while working as part of the overall system.
Example: In the purchasing concierge system, burger ordering and pizza ordering are handled by separate agents.
Benefits:

- Easier maintenance
- Clear separation of responsibilities
- Independent component development
- Flexible integration

The lecture also draws an analogy with the Mixture of Experts (MoE) approach, where specialized components handle different aspects of a task.

4. Tutorial Objectives and Technologies

The lecture introduces an interoperable, open-source-oriented AI agent architecture.
Main technologies covered:

Technology	Purpose in the lecture
CrewAI	Building the Burger Seller Agent
LangGraph	Building the Pizza Seller Agent
Google ADK	Building the Route Agent
Agent-to-Agent (A2A) Protocol	Communication and interoperability between agents
LiteLLM	Accessing different model providers through a common interface

Goal: Build a system in which agents developed with different frameworks can cooperate.

5. Open and Decentralized AI Technology Stack

Definition: An open and decentralized AI stack uses independently developed components and open technologies to create a flexible AI system.

The lecture's architecture includes:

- A main Route Agent
- Supporting specialized agents
- Communication protocols
- AI models
- Hardware and runtime infrastructure

Key Point: Different components can work together without requiring every agent to use the same framework.

6. Why Run Your Own AI Stack?

The lecture identifies several reasons to operate and control your own AI infrastructure.

A. Avoid Vendor Lock-In

Reduce dependence on a single provider by designing a system that can work with alternative providers and components.

B. Data Privacy

Maintain greater control over where data is stored and processed.

C. Deep Customization

Customize agents, workflows, tools, and models according to requirements.

D. Cost Optimization

Select models and infrastructure based on cost and performance.

E. Performance Optimization

Choose suitable models and computing resources for different tasks.

Key Point: Running your own stack provides greater control, but requires infrastructure, maintenance, and security management.

7. Purchasing Concierge System — Main Example

Definition: A purchasing concierge is an assistant that helps users

complete purchasing tasks by coordinating specialized services.
The lecture uses a purchasing concierge system to demonstrate multi-agent collaboration.

Components

Agent	Responsibility
Route Agent	Directs the request to the appropriate specialized agent
Burger Seller Agent	Handles burger-related requests
Pizza Seller Agent	Handles pizza-related requests

Example Workflow

1. A user submits a food order request.
2. The Route Agent identifies which specialized agent should handle it.
3. The Burger Agent or Pizza Agent processes the relevant request.
4. Agents communicate as needed to complete their assigned work.
5. The system returns the result to the user.

Key Point: The Route Agent coordinates the workflow, while the specialized agents handle their respective tasks.

8. Fundamental Building Blocks of Agent Systems

A. AI Agent

A system that uses a model, instructions, and potentially tools to perform tasks.

B. Agent Card

A machine-readable description of an agent's identity and capabilities. In A2A-based systems, it can help other agents discover what an agent offers and how to communicate with it.

C. Agent-to-Agent (A2A) Protocol

A communication protocol designed to allow independently implemented AI agents to interact and collaborate.

D. Agent Roles

Each agent has a specific responsibility, such as routing requests, selling burgers, or selling pizzas.

E. Communication

Agents exchange requests, information, and results to coordinate their tasks.

F. Multi-Agent Workflow

A sequence of activities in which agents perform specialized tasks and work toward a shared outcome.

Example: The Route Agent directs a burger request to the Burger Seller Agent.

9. Managing Agent Context: Session, State, and Memory

These concepts help an agent manage information during interactions.

A. Session

Definition: A session represents a particular interaction or conversation.
Purpose: Helps maintain continuity during an ongoing interaction.

B. State

Definition: State contains information about the current progress or condition of a task or workflow.
Purpose: Helps an agent determine what has already happened and what remains to be done.

C. Memory

Definition: Memory stores information that may be useful beyond the current step or interaction.
Purpose: Helps retain relevant information for later use, depending on the system's memory design.

Comparison

Concept	Main Purpose
Session	Maintains interaction continuity
State	Tracks current workflow information
Memory	Retains useful information for future use

Key Point: Session, state, and memory are related, but they serve different purposes.

10. Security and Governance in Agent Systems

Multi-agent systems need security and operational controls to work safely and reliably.

Important Elements

- **HTTPS:** Protects data in transit when correctly configured.
- **Authentication:** Verifies the identity of a user, service, or agent.
- **Authorization:** Determines which resources or actions an authenticated identity may access.
- **Observability:** Helps monitor system activity and performance.
- **Debugging:** Helps identify and resolve problems.
- **API Management:** Controls how services and interfaces are exposed and used.
- **Agent Onboarding:** Registers and configures agents so they can participate in the system.

Key Point: Security, monitoring, and access control are essential when agents communicate and perform actions.

11. Building the Burger Seller Agent with CrewAI

Definition: CrewAI is a framework for building AI agent systems that can coordinate roles and tasks.
The lecture demonstrates constructing the Burger Seller Agent using CrewAI.

Main Steps

1. Define the agent's role.
2. Configure its tools.
3. Integrate a Large Language Model (LLM).
4. Define how the agent handles its task.
5. Expose the agent's capabilities for communication.
6. Initialize the server.

Example: The Burger Seller Agent handles burger-related requests within the purchasing concierge system.

12. Building the Pizza Seller Agent with LangGraph

Definition: LangGraph is a framework for building stateful, often multi-step AI workflows using graphs.

The lecture uses LangGraph to implement the Pizza Seller Agent.

Main Steps

1. Assign the model.
2. Represent the agent and its workflow.
3. Configure its internal reasoning and action loop.
4. Execute the required actions.
5. Deploy the agent.

Example: The Pizza Seller Agent handles pizza-related requests and works as part of the larger concierge system.

13. Building the Route Agent with Google ADK

Definition: Google Agent Development Kit (ADK) is a framework for developing and orchestrating AI agents.

The lecture uses Google ADK to build the Route Agent.

Main Steps

1. Integrate the language model.
2. Use LiteLLM for model-provider access.
3. Configure the ADK runtime and event loop.
4. Orchestrate requests between the specialized agents.
5. Return the final result through the system.

Example: The Route Agent directs food-order requests to the Burger Seller Agent or Pizza Seller Agent.

14. Complete Multi-Agent System Workflow

The final demonstration brings the components together.

Overall Workflow

User Request → Route Agent → Specialized Agent → Result → User

For example:

- A burger request is directed to the Burger Seller Agent.
- A pizza request is directed to the Pizza Seller Agent.
- The relevant agent handles its assigned task and returns a result through the overall workflow.

The agents use their configured frameworks, models, and communication

mechanisms to collaborate.
Purpose: Demonstrate that agents built using different technologies can participate in one coordinated system.

Quick Revision Sheet

Term	Meaning
AI Agent	A system that uses AI and possibly tools to perform tasks
Multi-Agent System	Multiple agents cooperating on a larger task
Modularity	Dividing a system into specialized components
A2A Protocol	Communication between agents
Agent Card	Description of an agent's identity and capabilities
Route Agent	Directs requests to the appropriate agent
Session	Represents an interaction
State	Tracks current task or workflow information
Memory	Stores information for later use
Authentication	Verifies identity
Authorization	Controls access permissions
CrewAI	Framework used for the Burger Seller Agent
LangGraph	Framework used for the Pizza Seller Agent
Google ADK	Framework used for the Route Agent
LiteLLM	Provides a common interface to model providers

Final Takeaway

The lecture demonstrates how to build an interoperable multi-agent system using specialized burger, pizza, and routing agents. CrewAI, LangGraph, and Google ADK serve different agent roles, while the A2A Protocol supports communication between agents. Session, state, memory, and security controls help make the system more manageable and reliable.